Peer Review: "The Geometry of Thought: A Hilbert Space Analysis of Semantic Categories Reveals Two Basins of Attraction in the SGP Topographic Field"

Reviewer: Anonymous
Submission to: eLife
Recommendation: Reject with invitation to resubmit
Date: April 2026

Executive Summary

This manuscript presents an application of the Sentient-Field Hypothesis / Sentient-Generative Principal (SFH-SGP) framework to analyze semantic category organization in TRIBE v2 foundation model predictions. While the work demonstrates technical competence in data analysis and makes an interesting attempt at bridging theoretical neuroscience with computational methods, it suffers from fundamental empirical and conceptual limitations that prevent acceptance at a venue like eLife.

Primary concerns:

1. **No empirical validation**: Analysis relies entirely on foundation model predictions without validation against real fMRI data
2. **Circular reasoning in framework design**: SFH-SGP primitives are derived from the same data used to validate them
3. **Theoretical framework not independently established**: The SFH-SGP model is not well-grounded in established neuroscience
4. **Overstated claims**: 100% LOO-CV accuracy suggests model overfitting to 10 semantic categories
5. **Limited scope**: 10 categories with 91–107 samples each is insufficient for robust claims about semantic organization
6. **Missing mechanistic insight**: The work describes geometry but offers limited explanation for *why* this structure exists

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Detailed Comments

1. MAJOR ISSUES

1.1 Absence of Empirical Validation Against Real fMRI Data

Issue: The entire analysis rests on TRIBE v2 model predictions, not actual neuroimaging. While the authors acknowledge this in limitations (Section 4.4), the severity is understated.

Problem:

- Foundation models are trained to *match* observed patterns in their training data but can introduce artifacts, biases, or spurious structure that doesn't reflect neurobiology
- TRIBE v2 was trained on 720 subjects' fMRI data, but it is not clear whether the 1,022 text stimuli in this analysis overlap with that training set, introducing potential generalization concerns
- No cross-validation against held-out real fMRI data is performed

Recommendation:

- Validate the basin structure against independent fMRI data (e.g., from the Human Connectome Project, Berkeley Sentence Corpus fMRI, or similar open datasets)
- If basin separation holds in real fMRI, statistical effect sizes should be reported
- Explicitly discuss potential differences between predicted and observed activation patterns

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1.2 Circular Validation of the SFH-SGP Framework

Issue: The SFH-SGP primitives (C, F, χ) are constructed from the same differential activation matrix used to test the framework's predictions.

Problem:

- Coherence (C) is defined as "projection onto the leading eigenvector of the differential co-activation matrix" (Section 2.3, item 4)—this is derived from the data being analyzed
- Fertility (F) is operationalized as G5_dmn differential activation
- The Sentient Potential $\chi = \alpha C + \beta F$ is then used to classify categories and demonstrate basin separation
- This is a form of circular reasoning: the framework defines the structure it then claims to reveal

Specific concern:

- If you define a functional that emphasizes certain nodes, those nodes will naturally show high within-category coherence. This is tautological, not predictive.
- For the framework to be validated, it must make *a priori* predictions about basin structure before analyzing the data

Recommendation:

- Split the data: use half to define C and F from first principles (without knowledge of category membership), then test predictions on the held-out half
- Alternatively, provide strong theoretical justification (separate from data) for why Coherence and Fertility should organize categories as observed
- Show that simpler baselines (e.g., clustering on raw differential activation, without χ weighting) perform worse

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1.3 Overfit Classification Performance (100% LOO-CV Accuracy)

Issue: Leave-One-Out Cross-Validation achieves perfect (100%) classification accuracy for basin membership.

Red flag:

- With only 10 categories and 1,022 samples total (~100 per category), 100% LOO-CV is unrealistic unless:
  - The categories are completely separable in the feature space (suggesting the categories themselves, not the framework, are driving the result)
  - The classifier is overfitting to noise or artifacts in the model predictions

Missing analysis:

- No confusion matrix shown for LOO-CV (Figure 4 shows a bar chart, not detailed errors)
- No comparison to null models (e.g., random assignment, permutation testing)
- No analysis of which samples are most/least confidently classified
- No sensitivity analysis: what happens if you perturb α or β?

Recommendation:

- Provide the full confusion matrix with per-category accuracy
- Test performance with simpler baselines (e.g., k-NN on raw differential activation, SVM without χ weighting)
- Conduct permutation testing: shuffle category labels and refit; basin separation should disappear
- Report effect sizes for basin separation with confidence intervals

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1.4 Theoretical Framework Lacks Independent Grounding

Issue: The SFH-SGP framework is presented as established fact, but its theoretical foundations are not independently validated in the peer-reviewed literature.

Concerns:

- The framework references "Traver, 2026b" and "Traver, 2026a" (presumably preprints or unpublished work)
- eLife expects submission of established methodologies or strong theoretical justification for novel ones
- The claim that consciousness is "the fundamental fabric of reality" (per memory context) is metaphysical and not empirically testable without operational definitions
- The paper does not engage with competing frameworks (e.g., predictive coding, free energy principle, competing basin/attractor models in neuroscience)

Specific issue:

- The dual-stream interpretation (Section 3.4, 4.2) invokes Hickok & Poeppel (2007), but the connection is superficial. The paper shows that basins have opposite patterns across streams but doesn't mechanistically explain why or connect to the underlying linguistics/motor control theories

Recommendation:

- Situate SFH-SGP within existing theories of semantic representation (e.g., experiential grounding, semantic hubs, dual-process theories)
- Provide mathematical justification for the specific form of $\chi = \alpha C + \beta F$ (why not other combinations? why equal weights?)
- Discuss competing explanations for the observed basins (e.g., sampling bias, stimulus properties, artifacts in TRIBE v2)

1.5 Limited Generalization: 10 Categories Insufficient for Claims About Semantic Organization

Issue: The analysis covers only 10 semantic categories with ~100 samples each.

Problem:

- Modern semantic space studies (e.g., Huth et al. 2016; covered here but not fully engaged) use 1,000+ stimuli spanning continuous semantic dimensions
- 10 discrete categories are not representative of the full landscape of human conceptual knowledge
- The claimed binary split (embodied vs. abstract) is intuitive but reductive—many categories lie on spectra (e.g., "social" involves both embodied and abstract elements)
- Sample size per category is modest; statistical power for detecting subtle basin structure is limited

Specific concern:

- The paper hypothesizes that "motor, auditory, emotional, memory, simple, social" cluster together (Section 1.3), but these are qualitatively diverse. Why does memory (internal, temporal) cluster with auditory (external, sensory)? The paper doesn't justify this grouping *a priori*

Recommendation:

- Expand to 50+ categories or use a continuous semantic space (e.g., word embeddings across a large lexicon)
- Test whether the two-basin structure is robust to different category groupings or merging/splitting categories
- Analyze fine-grained basin structure: are there subclusters within basins? Do basins overlap at boundaries?

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2. MODERATE ISSUES

2.1 Methods Section Lacks Critical Detail

Data and Code Availability:

- The paper cites "Traver, 2026a" for "complete analysis pipeline and code," but no repository link is provided in the text
- The TRIBE v2 model details are sparse; no code for accessing TRIBE predictions is provided
- For reproducibility, all code, data (or instructions to generate it), and numerical results should be in supplementary materials or a public repository

Statistical Analysis:

- No mention of multiple comparison correction for the nine ANOVAs per node (Section 3.1)
- Effect sizes (Cohen's d) are provided in Table 3 but not consistently reported; what is α for statistical significance?
- Silhouette analysis mentioned in methods (Section 2.4) but not reported in results
- No discussion of hierarchical clustering method selection (Ward's vs. single/complete linkage)

Recommendation:

- Provide explicit GitHub/Zenodo links and DOI for reproducibility
- Report p-values and 95% confidence intervals for all main results
- Apply Bonferroni or FDR correction to multi-test analyses
- Include full silhouette analysis in results or supplementary material

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2.2 The χ Functional Is Not Well Motivated

Mathematical Form:

- Why is χ a *linear* combination of C and F? No justification is provided
- Why equal weights ($\alpha = \beta = 1.0$)? Sensitivity analysis should show robustness to other values
- The Langevin equation (Eq. 1.2) is presented but never used in the analysis; its role is unclear

Biological Interpretation:

- What does maximizing/minimizing χ mean neurobiologically? The paper states "the system evolves toward critical points where $\nabla\chi = 0$—the Resonance Anchors" but offers no mechanistic interpretation
- Why should Coherence and Fertility be the only dimensions? What about energy consumption, temporal dynamics, or other constraints?

Recommendation:

- Derive χ from first principles (e.g., information-theoretic optimality, constraint satisfaction)
- Show that the specific functional form outperforms alternatives (e.g., multiplicative combination, polynomial terms)
- Conduct sensitivity analysis: vary α and β, show that basin structure is robust
- Provide biological interpretation of what "Resonance Anchors" are (attractor states? stable fixed points? something else?)

2.3 The "Fertility" Metric Is Opaque

Definition:

- Fertility F is operationalized as G5_dmn differential activation (Section 2.3, item 3)
- But Table 3 shows that G5_dmn has Cohen's d = 0.54 (smallest effect) and does not significantly distinguish basins
- This contradicts the importance ascribed to Fertility in the framework

Concern:

- Why should the Default Mode Network's activity specifically represent "generative potential"? This is asserted without justification
- The DMN is typically active during rest and self-referential thought, not during semantic processing of presented stimuli
- If DMN differential is not significant, how does it contribute meaningfully to χ?

Recommendation:

- Provide independent justification for why DMN = Fertility (e.g., literature review, theoretical argument)
- If DMN differential is not significant, consider removing it or replacing with a more predictive metric
- Test alternative operationalizations of "generative potential" (e.g., posterior probability in language models, entropy of predicted continuations)

2.4 Interpretation of Stream Analysis Is Speculative

Stream-Basin Relationship (Section 3.4, 4.2):

- The paper claims that "both basins show opposing patterns across all streams" (Figure 3, Table 3)
- But this is partly an artifact of how streams are defined: if you partition nodes into streams and then find differential activation, opposing patterns may emerge naturally
- The connection to Hickok & Poeppel (2007) is superficial; the paper doesn't explain *why* embodied vs. abstract cognition should engage dorsal/ventral streams oppositely

Mechanistic gap:

- The paper states: "embodied processing involves suppression of executive and abstract systems, while abstract processing involves suppression of sensory-motor systems" (Section 4.2)
- But this is inferred from differential activation patterns, not from causal manipulations or mechanistic modeling
- No evidence is provided that this is suppression (inhibition) vs. decoupling vs. simple resource allocation

Recommendation:

- Be more cautious in causal claims; describe observed patterns without implying mechanisms
- Test whether stream patterns are robust to alternative parcellation schemes (e.g., other atlases)
- Engage more deeply with the dual-stream literature to explain observed patterns within existing theory

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3. MINOR ISSUES

3.1 Presentation and Clarity

Figures:

- Figure 1 (differential activation heatmap) is hard to interpret; categories are sorted by basin, making within-category heterogeneity unclear
- Figure 2 (right panel) would benefit from individual data points, not just error bars
- Figure 5 (3D landscape) is visually interesting but difficult to interpret; a 2D projection with contours might be clearer

Text:

- Abstract is dense; consider simplifying the methods paragraph for accessibility
- The paper uses "Sentient Potential" throughout, which has philosophical connotations not well-grounded in neurobiology; "activation potential" or "semantic potential" might be clearer
- Key concepts (Coherence, Fertility, Quota) are introduced but not consistently explained; a glossary would help

Recommendation:

- Redesign Figure 1 to sort categories by χ value and show individual stimulus effects
- Add violin plots to Figure 2 showing per-category distributions
- Simplify abstract and add a one-sentence summary of methods

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3.2 References and Citations

Issues:

- Heavily self-cited (Traver 2026a, 2026b); while self-citation is acceptable for methodology, the framework references are to unpublished work
- Missing recent work on semantic organization (e.g., Huth et al. 2016, cited; but also more recent studies on semantic manifolds, embedding spaces)
- The TRIBE v2 reference is to a Meta blog post, not a peer-reviewed publication; arXiv preprint reference is provided (d'Ascoli et al., 2025), but the 2026 "technical report" is vague

Recommendation:

- Provide proper citations for all framework publications (or preprints on arXiv with DOI)
- Expand related work section to situate findings within semantic cognition literature
- If citing blog posts, also include the peer-reviewed preprint

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3.3 Missing Analyses

Robustness:

- No leave-one-category-out cross-validation (remove entire category, test prediction)
- No sensitivity analysis: what if you merge similar categories or split heterogeneous ones?
- No test of generalization to new categories not in the training analysis
- No analysis of individual differences: do all subjects show the same basin structure in real fMRI?

Neurobiological validation:

- No comparison to published semantic maps (e.g., Huth et al. 2016)
- No analysis of how basin structure relates to semantic similarity (e.g., do semantically similar categories cluster within basins?)
- No investigation of whether basins correspond to known functional hierarchies in the brain

Recommendation:

- Conduct leave-one-category-out CV to assess true generalization
- Compare observed basin structure to published semantic atlases
- Test whether basins predict semantic distances between categories

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4. MISSING OR UNDERDEVELOPED SECTIONS

4.1 Related Work

The paper lacks a substantial "Related Work" or "Background" section that would:

- Review competing theories of semantic organization (grounded cognition, semantic hubs, dual-process models)
- Position SFH-SGP within existing frameworks
- Discuss prior basin/attractor models in neuroscience
- Explain why foundation models like TRIBE v2 are appropriate for testing these theories

Recommendation:

- Add a 1–2 page "Related Work" section before Methods
- Clearly state what is novel about the SFH-SGP approach vs. prior work

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4.2 Alternative Explanations

The paper does not adequately address whether the observed two-basin structure could arise from:

- **Stimulus properties**: Are embodied categories simply "simpler" stimuli? Does stimulus word frequency, length, or semantic richness explain basin membership?
- **Embedding geometry**: Does the tinyllama embedding space show the same bipartition? If so, the structure is pre-existing in the language model, not revealed by fMRI encoding
- **TRIBE v2 artifacts**: Could the two-basin structure be an artifact of TRIBE's training procedure or architecture?

Recommendation:

- Analyze stimulus properties (word frequency, length, concreteness ratings) and their relationship to χ and basin membership
- Extract embeddings from tinyllama and test whether two-basin structure is present before fMRI encoding
- Compare TRIBE v2 predictions to simpler encoding models (e.g., linear regression on embeddings) to assess whether TRIBE-specific structure is necessary

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5. SUGGESTIONS FOR RESUBMISSION

If the authors plan to revise and resubmit to eLife or another tier-1 venue, I recommend addressing the following in order of importance:

1. **Empirical Validation (Essential):**
    - Validate basin structure against real fMRI data (any open dataset will suffice)
    - Show that the two-basin separation holds in empirical data and that effect sizes are reasonable
2. **Theoretical Grounding (Essential):**
    - Provide independent justification for SFH-SGP framework or clearly position it as a novel hypothesis requiring validation
    - Avoid circular reasoning: separate hypothesis generation from hypothesis testing
3. **Expanded Scope (Important):**
    - Extend to ⩾50 semantic categories or a continuous semantic space
    - Test generalization to new categories and stimuli not used in analysis
4. **Mechanistic Insight (Important):**
    - Move beyond describing geometry to explaining mechanisms
    - Why does the brain organize semantics this way? What are the computational advantages?
5. **Robustness and Transparency (Important):**
    - Provide full reproducibility: code, data, and detailed statistical reporting
    - Conduct sensitivity analyses for all key parameters
    - Discuss alternative explanations for observed patterns

Minor Suggestions

- Consider renaming "Sentient Potential" to avoid philosophical connotations
- Add supplementary material with: full confusion matrices, silhouette analysis, sensitivity plots, alternative hierarchical clustering dendrograms
- Include a schematic diagram of the SFH-SGP framework (conceptual model) before mathematical introduction
- Use consistent terminology: "Real-World" vs. "Embodied" (used interchangeably)

Verdict

This manuscript demonstrates technical competence and tackles an interesting problem—understanding the geometric organization of semantic categories in neural activation space. However, it is not ready for publication at eLife due to reliance on unvalidated model predictions, circular framework validation, and insufficient empirical grounding. The two-basin structure is intriguing, but the evidence is insufficient to support strong claims about how the brain organizes semantic knowledge.

With substantial revisions addressing the major issues above—particularly empirical validation against real fMRI data and decoupling hypothesis generation from hypothesis testing—this could become a solid contribution to computational neuroscience.

Recommendation

REJECT with invitation to resubmit after major revisions.

The work would benefit from:

1. Validation against real fMRI data
2. Removal of circular reasoning in framework design
3. Expanded scope and generalization tests
4. More careful mechanistic interpretation

Reviewer Confidence: High (domain expertise in computational neuroscience, neuroimaging, and semantic cognition)